

FACILITY ONBOARDING TEMPLATE

Compound Risk Detection Engine — Add Facility Document

Facility Name: Riverbend Thermal Power Station

Prepared By: Meera Iyer

Designation: Deputy Plant Head - Operations

Date Prepared: June 15, 2026

1. Facility Overview

Facility / Plant Name: Riverbend Thermal Power Station

Physical Address: NH-49, Riverbend Industrial Corridor, Raigarh District, Chhattisgarh, India

Industry Sector: Thermal Power (Coal-Fired)

Operating Company: Riverbend Power Generation Ltd.

Commissioning Date: March 15, 2025

Installed Capacity: 600 MW (Unit 1, sub-critical boiler)

Major departments/units:

Department / Unit Name	Primary Function	Approx. Headcount
Coal Handling Plant	Receiving, crushing, and conveying coal to bunkers	45
Boiler & Furnace Operations	Combustion and steam generation	38
Turbine-Generator	Power generation from high-pressure steam	22
ESP & Ash Handling	Emission control and ash disposal	18
Maintenance	Mechanical, electrical, and instrumentation upkeep	52
Safety & Security	Plant safety, fire response, and access control	20

Narrative overview:

Riverbend Thermal Power Station is a 600 MW sub-critical coal-fired power plant supplying electricity to the regional grid. Coal is received by rail and road, crushed and conveyed to bunkers, then pulverized and fired into a single boiler unit. Steam drives a single turbine-generator set. Flue gas passes through an electrostatic precipitator before discharge. The plant reached commercial operation in March 2025 following a multi-year construction delay and is currently in its second full year of continuous operation.

2. Process Flow Description

Coal arrives by rail and is unloaded via wagon tipplers into the coal handling plant, where it is crushed to the required size and conveyed to day bunkers via a network of belt conveyors. From the bunkers, coal is fed through gravimetric feeders into coal mills (pulverizers), which grind it to a fine powder and transport it pneumatically into the furnace via the Primary Air (PA) fan system. Secondary Air (SA), supplied separately, completes the combustion air requirement. Combustion in the furnace generates heat that converts feedwater circulating through the boiler's water-wall tubes into steam, which is further superheated before being routed to the turbine. The turbine drives the generator to produce electrical power, which is stepped up via the generator transformer and exported to the grid. Exhaust steam from the turbine is condensed in a surface condenser cooled by the circulating water system and returned to the boiler as feedwater, closing the steam-water cycle. Flue gas exiting the furnace passes through the Electrostatic Precipitator (ESP) for particulate removal before being drawn out by the Induced Draft (ID) fan and released via the stack. Bottom ash and fly ash are collected separately and conveyed to the ash handling plant for onward disposal or sale.

Referenced Drawing Numbers: PFD-RTPS-001 (Process Flow Diagram), PID-RTPS-BLR-014 (Boiler P&ID)

3. SCADA / DCS Systems

System Name	Vendor	Version	Primary Function	Redundant? (Y/N)
Unit 1 DCS	ABB Ability Symphony Plus	v3.2	Overall unit control and monitoring	Y
Boiler Protection System	Siemens SPPA-T3000	v8.1	Master Fuel Trip and boiler interlocks	Y
Turbine Governor System	GE Mark VIe	v6.0	Turbine speed and load control	Y

Historian System: OSIsoft PI System

Network Architecture Notes: IT/OT segregated via firewall DMZ; SCADA network is air-gapped from corporate IT with no direct internet route

4. Sensor & Instrumentation Details

Tag ID	Instrument Name	Location	Type	Unit	Normal Range	Alarm Threshold
PAF-01	Primary Air Fan Status	Boiler Unit 1 - Air Supply	Digital Status (On/Off)	-	Running	Fault / Tripped
SAF-02	Secondary Air Fan Status	Boiler Unit 1 - Air Supply	Digital Status (On/Off)	-	Running	Fault / Tripped
FPT-101	Furnace Draft Pressure Transmitter	Boiler Unit 1 - Furnace	Pressure	mm WC	-5 to +5	> +10 or < -10
TT-204	Superheater Outlet Temperature	Boiler Unit 1 - Superheater Header	Temperature	°C	480 - 540	> 560
CFR-05	Coal Feed Rate	Coal Mill Feeder	Flow Rate	t/hr	18 - 22	> 25
STP-301	Main Steam Line Pressure	Turbine Inlet	Pressure	bar	165 - 170	> 175
VIB-12	ID Fan Bearing Vibration	Induced Draft Fan	Vibration	mm/s	0 - 4.5	> 7.1
LDR-01	Generator Load	Turbine-Generator	Other	MW	550 - 600	Ramp rate > 20 MW/min
ASH-08	Bottom Ash Hopper Level	Boiler Unit 1 - Bottom Ash Hopper	Level	%	20 - 70	> 85
CO-15	Flue Gas CO Concentration	ESP Inlet Duct	Gas Concentration	ppm	0 - 150	> 300

4.1 Real-Time Data Integration (API Access)

API Base URL: <https://scada.riverbendpower.example.com/api/v1/tags>

Authentication Method: Bearer Token (rotated monthly)

Polling Frequency: Every 15 seconds

5. Key Personnel

5.1 Managerial Staff

Name	Designation	Department	Email	Phone
R. Venkatesh	Plant Head	Management	r.venkatesh@riverbendpower.example.com	+91 98765 10001
Meera Iyer	Deputy Plant Head - Operations	Management	meera.iyer@riverbendpower.example.com	+91 98765 10002
Suresh Bansal	Head of Maintenance	Maintenance	suresh.bansal@riverbendpower.example.com	+91 98765 10003
Kavita Rao	Chief Safety Officer	Safety	kavita.rao@riverbendpower.example.com	+91 98765 10004

5.2 Maintenance Staff

Name	Role	Shift	Email	Phone	Certifications
Ravi Kumar	Shift Engineer	Shift A	ravi.kumar@riverbendpower.example.com	+91 98765 43210	Boiler Operations Certified
Anil Sharma	Maintenance Technician	Day Shift	anil.sharma@riverbendpower.example.com	+91 98765 43211	Rotating Equipment Certified
Deepak Verma	Instrumentation Technician	Day Shift	deepak.verma@riverbendpower.example.com	+91 98765 43213	SCADA/DCS Certified
Sanjay Patel	Electrical Technician	Shift B	sanjay.patel@riverbendpower.example.com	+91 98765 43214	HV Systems Certified

5.3 Safety Officers

Name	Role	Email	Phone
Priya Nair	Safety Officer	priya.nair@riverbendpower.example.com	+91 98765 43212
Arjun Mehta	Fire & Emergency Response Officer	arjun.mehta@riverbendpower.example.com	+91 98765 43215

5.4 Escalation Logic

Level	Trigger Condition	Contact / Role	Response Time Expected
1 — First Responder	Sensor faults, gas/pressure alarms	On-duty Shift Engineer	Immediate (on-site)
2 — Shift Engineer	Level 1 unresolved within 10 minutes	Ravi Kumar (or on-shift equivalent)	Within 10 minutes
3 — Plant Head	Level 2 unresolved within 30 minutes, or safety-critical	R. Venkatesh	Within 30 minutes
4 — Corporate / Regulatory	Injury, fire, or reportable regulatory event	Corporate Safety Cell + Chief Inspector of Factories	Within 2 hours

6. Shift & Workforce Patterns

Shift Name	Start Time	End Time	Typical Headcount
Shift A	6:00 AM	2:00 PM	65
Shift B	2:00 PM	10:00 PM	60
Shift C	10:00 PM	6:00 AM	40

Shift changeover follows a documented handover checklist covering equipment status, active alarms, open permits, and any deferred maintenance items. Outgoing and incoming shift engineers overlap for 15 minutes to walk the checklist together and sign a shared logbook. Note: Primary Air Fan (PAF-01) preventive maintenance windows are typically scheduled to begin at 2:00 PM to minimize disruption to Shift A production targets -- this coincides directly with the Shift A to Shift B changeover window.

7. Maintenance Records & Timelines

Equipment	Last Maintenance	Type	Next Scheduled	Performed By	Notes
Primary Air Fan (PAF-01)	2026-04-10	Preventive (bearing lubrication)	2026-07-10	Anil Sharma	On schedule
Induced Draft (ID) Fan	2026-03-22	Preventive (vibration check)	2026-06-22	VibroTech (external contractor)	Overdue by 15 days as of filing
Boiler Tube Inspection	2025-11-05	Statutory Inspection	2026-11-05	Chief Boiler Inspector's Office	—
Coal Feeder Belt	2026-05-18	Preventive (belt tension)	2026-08-18	Anil Sharma	—
Fire Water Pump	2026-01-30	Preventive (test run)	2026-07-30	Arjun Mehta	Deferred once already from April

8. Incident & Negligence History

Date	Incident Description	Root Cause	Corrective Action	Recurred ?
2025-08-14	PA Fan bearing overheating detected during routine round	Missed lubrication interval	Immediate lubrication, bearing replaced	N
2025-11-02	Near-miss: contractor entered ash hopper area without permit	Communication gap at shift handover	Retrained contractor supervisors on PTW process	Y
2026-02-20	Same permit-to-work violation recurred (ash hopper)	Level 1 escalation contact on leave, no backup assigned	Assigned backup escalation contact	N since

Broader negligence history:

In 2025, an ID Fan vibration alarm was acknowledged by the shift engineer but the corresponding maintenance action was deferred twice due to contractor scheduling conflicts before finally being addressed. No injuries resulted. A corporate safety audit in November 2025 separately flagged the plant's escalation backup coverage as a gap (no designated backup when a Level 1 contact is on leave) -- this has since been addressed as of March 2026, following the February 2026 recurrence noted above.

9. Employee Directory

Name	Role / Dept.	Email	Phone	Blood Group	Working Hours	Working Days	Emergency Contact
Ravi Kumar	Shift Engineer / Boiler Ops	ravi.kumar@riverbendpower.example.com	+91 98765 43210	O+	6AM-2PM	Rotating Shift A	Sunita Kumar / +91 98765 11111 / Spouse
Anil Sharma	Maint. Tech / Maintenance	anil.sharma@riverbendpower.example.com	+91 98765 43211	B+	9AM-6PM	Mon-Sat	Rakesh Sharma / +91 98765 22222 / Brother
Priya Nair	Safety Officer / Safety	priya.nair@riverbendpower.example.com	+91 98765 43212	A-	8AM-5PM	Mon-Fri	Vijay Nair / +91 98765 33333 / Spouse
Deepak Verma	Instr. Tech / I&C	deepak.verma@riverbendpower.example.com	+91 98765 43213	AB+	9AM-6PM	Mon-Sat	Neha Verma / +91 98765 44444 / Spouse
Sanjay Patel	Elec. Tech / Electrical	sanjay.patel@riverbendpower.example.com	+91 98765 43214	O-	2PM-10PM	Rotating Shift B	Meena Patel / +91 98765 55555 / Mother
Arjun Mehta	Fire & ER Officer / Safety	arjun.mehta@riverbendpower.example.com	+91 98765 43215	B-	On-call rotation	Rotating	Kiran Mehta / +91 98765 66666 / Father
Lakshmi Reddy	Control Room Op. / Boiler Ops	lakshmi.reddy@riverbendpower.example.com	+91 98765 43216	A+	10PM-6AM	Rotating Shift C	Ganesh Reddy / +91 98765 77777 / Spouse
Mohit Joshi	Coal Handling Supv. / Coal Handling	mohit.joshi@riverbendpower.example.com	+91 98765 43217	AB-	6AM-2PM	Rotating Shift A	Pooja Joshi / +91 98765 88888 / Spouse

10. Attendance & Access Control Systems

System Name / Vendor: BioTrack Enterprise Attendance & Access Control

API Base URL: <https://attendance.riverbendpower.example.com/api/status>

Authentication Method: API Key

Data Format: JSON

11. Utility & Support Systems

Utility System	Type / Vendor	Backup / Redundancy	Last Tested Date
Emergency Power Backup	2x 1500 kVA DG Sets	N+1 redundancy	2026-06-01
Compressed Air System	Atlas Copco GA-90 x2	N+1 redundancy	2026-05-15
Water Treatment / Cooling Water	Cooling tower + DM plant	Single-train, no redundancy	2026-04-20
Fire Water System	Dedicated ring main + jockey pump	N+1 pumps	2026-01-30
HVAC (Control Room)	Dual redundant AHUs	N+1 redundancy	2026-03-10

12. Training & Certification Records

Safety Induction Program: 2-day program for all new hires; annual refresher for existing staff

Mock Drill Frequency: Quarterly (fire); Semi-annual (gas leak / evacuation)

Date of Last Mock Drill: April 18, 2026 (fire evacuation)

Contractor Safety Training: Mandatory 4-hour orientation before site access; logged in contractor management system

13. Environmental & Quality Compliance

Environmental Clearance: EC/CG/RTPS/2024/0187

Pollution Control Board Registration: CGPCB/CTO/2025/0456

Quality Management System Certification: ISO 9001:2015

Environmental Management System Certification: ISO 14001:2015

Occupational Health & Safety Certification: ISO 45001:2018

14. Emergency Response & Compliance

Regulatory Standards: Factories Act 1948, CEA Technical Standards for Construction, Boilers Act 2025, State Pollution Control Board norms

Fire Safety Systems Present: Hydrant network, turbine hall sprinklers, heat/smoke detection in control room and cable galleries, dedicated fire water ring main

Date of Last Safety Audit: November 12, 2025 (Corporate Safety Cell)

Key findings from last safety audit:

Overall compliance rated satisfactory. Two observations were raised: (1) an escalation backup coverage gap during personnel leave, since resolved as of March 2026; (2) a recommendation to increase ID Fan vibration monitoring frequency given a history of deferred maintenance on that equipment -- implementation is in progress.

Declaration

I confirm that the information provided in this document is accurate and complete to the best of my knowledge at the time of submission.

Prepared By: Meera Iyer, Deputy Plant Head - Operations

Reviewed By: R. Venkatesh, Plant Head

Date: June 15, 2026